YUVA Intern — Week 1 Task: Wireframe to Responsive Webpage

Intern Role: Frontend Developer Intern

Program: YUVA Intern

Assignment: Week 1 — Transforming a Wireframe Concept into a Complete, Responsive Static Webpage

Project Name: ApexHub — Sustainable Tech & Innovation Co-Working Space

Tech Stack: HTML5, CSS3, Vector SVG Assets, Headless Chrome / Playwright Testing Suite

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1. Introduction and Objective

The primary objective of the Week 1 internship task at YUVA Intern is to bridge the gap between UI/UX conceptual design and production-grade frontend implementation. As a frontend intern, the task requires independently conceptualizing a digital product, constructing a comprehensive low-fidelity wireframe representing both desktop and mobile viewports, and implementing the full design as a responsive, accessible static webpage using semantic HTML5 and modern CSS3—strictly without relying on external CSS frameworks (such as Bootstrap or Tailwind) or JavaScript libraries.

Key learning and execution goals for this project include:

- Establishing a clear visual hierarchy and layout structure.
- Translating low-fidelity wireframe blueprints into pixel-accurate HTML and CSS.
- Demonstrating semantic HTML elements (
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- Utilizing modern CSS layout engines (Flexbox and CSS Grid) for multi-dimensional alignment.
- Architecting a scalable CSS design system with CSS custom properties (variables), fluid typography (`clamp()`), and structured spacing tokens.
- Ensuring responsive adaptability across all four mandatory standard viewport ranges (Mobile 320–480px, Tablet 481–768px, Laptop 769–1024px, Desktop 1025px+).

- Validating the interface through automated browser rendering tests to guarantee zero unintended horizontal scrolling, complete asset loading, and full keyboard/accessibility compliance.

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2. Selected Website Concept and Reason for Selection

Chosen Concept

ApexHub: Next-Generation Sustainable Innovation & Co-Working Space

Tagline: "The Future-Ready Workspace for Modern Tech Innovators."

Rationale and Evaluation

During the initial concept evaluation phase, several typical options were considered, such as basic travel landing pages, generic personal portfolios, and standard e-commerce product pages. A clean-tech co-working and innovation hub was selected as the superior concept for the following reasons:

1. **Rich Structural Diversity:** A co-working platform requires multiple distinct layout patterns—a high-impact Hero with statistical callouts, a 6-card feature grid for technological amenities, an asymmetric workspace showcase with tiered cards, a 3-tier membership pricing comparison table, social proof testimonial quotes with avatars, an interactive tour booking form, and a multi-column semantic footer.

2. **Demonstration of Both Flexbox and CSS Grid:**

- **CSS Grid** is ideal for the 2D multi-column layouts: the 6-amenity grid (3 columns x 2 rows), the workspace showcase, and the 4-column footer.

- **Flexbox** is ideal for 1D directional alignment: navigation bars, button groups, badge pills, pricing checklist rows, testimonial author metadata, and horizontal metrics bars.

3. **Professional Relevance to Internships and Technology:** Co-working ecosystems resonate directly with modern remote, hybrid, and startup developer cultures. It provides a real-world scenario to display badges (e.g., "100% Solar-Powered", "Most Popular"), pricing tiers with monthly/annual cadences, and clear call-to-actions.

4. **Feasibility Without Complex JavaScript:** All functional requirements—including mobile hamburger navigation, interactive card elevation, hover transitions, focus rings, and section anchor scrolling—can be achieved using pure semantic HTML5 and CSS3.

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3. Wireframe Design and Layout Decisions

Before coding, an original low-fidelity vector wireframe was authored and saved to [`wireframe.svg`] (wireframe.svg). The wireframe establishes the layout blueprints side-by-side for both **Desktop (1024px+)** and **Mobile (375px–480px)** environments.

Desktop Layout Architecture (Left Artboard)

- **Top Bar (Sticky Header):** Height 70px. Left-aligned brand logo mark, centered horizontal navigation links (`Overview`, `Spaces`, `Amenities`, `Memberships`, `Stories`), and right-aligned "Book a Tour" primary action button.
- **Hero Section:** 2-column asymmetric layout (60/40 ratio). Left side features a sustainability badge, high-contrast

title, descriptive subhead, dual CTAs ("Explore Spaces" and "Schedule a Visit"), and a horizontal 3-metric statistics card. Right side features a modern architectural illustration of the workspace.

- **Amenities Grid:** Centered section header followed by a 3-column, 2-row CSS Grid displaying 6 key infrastructure features (Gigabit Fiber & AI Pods, Hardware/IoT Lab, Acoustic Booths, Biometric Access, Cafe, and Zen Lounge).
- **Flexible Workspaces Showcase:** 3-column card grid highlighting three core workspace offerings (Open Hot Desk, Dedicated Desk, and Executive Private Studio) with image headers, badge overlays, feature lists, pricing, and reservation buttons. The middle tier ("Dedicated Desk") is visually elevated as the recommended plan.
- **Membership & Pricing Tiers:** 3-column pricing matrix comparing Day Explorer ($25/day), Pro Resident ($350/mo), and Scale Enterprise ($890+/mo) with feature checklists.
- **Community Testimonials:** 3-column card layout displaying 5-star ratings, founder quotes, author avatars, names, and company credentials.
- **Tour Booking Section:** High-contrast dark container featuring a dual-column layout: left column contains visit perks and guarantees; right column contains an accessible form card with inputs for Full Name, Work Email, Desired Space, Hub Location, and Date.
- **Footer:** 4-column footer containing company mission and social icons, workspace links, ecosystem links, and an email newsletter subscription box.

Mobile Layout Architecture (Right Artboard)

- **Header:** Height 60px. Compact brand logo paired with an accessible 3-bar hamburger toggle.
- **Navigation:** Off-canvas / collapsible dropdown navigation menu triggered cleanly via CSS without external dependencies.

- **Hero Stacking:** Single-column vertical flow. The headline, copy, and CTAs stack naturally. Full-width buttons ensure effortless thumb tapping.
- **Grid-to-Stack Transformation:** The 3-column grids for amenities, workspace cards, pricing tiers, and testimonials transition smoothly into single-column vertical card stacks.
- **Touch-Target Sizing:** Form inputs, buttons, and navigation links maintain a minimum touch target of 44px to ensure seamless mobile usability.

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4. HTML Structure and Semantic Elements

The webpage is implemented inside [`index.html`](index.html). The markup adheres strictly to HTML5 validation standards and emphasizes document accessibility and semantic landmarks:

1. **Document Declarations & Metadata:**

- with .

- for complete character encoding.

- to ensure accurate responsive scaling.

- Descriptive and .

2. **Accessibility Features:**

- **Skip Link:** `Skip to main content` positioned at the very top of the DOM, hidden off-screen until keyboard focus is received.

- **Accessible Checkbox Navigation:** Pure CSS toggle using an ☐ coupled with .

- **Form Accessibility:** Every and